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Last login: Wed Jun 24 23:05:37 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern"
ls -R
adjacent_experiment          HalfShift_experiment
communication_comparison.csv  Mirrored_experiment
communication_comparison.png  plot_comparison.py
Comparison_terminal.pdf

./adjacent_experiment:
adjacent_communication.png
adjacent_graph.png
adjacent_latency.png
adjacent_sst.py
adjacent_stats.csv
Different Process count and SST_opCount .pdf
overall_scaling_graph.png
plot_scaling.py
plot_stats.py
Scaling
scaling_results.csv
Untitled.pdf

./adjacent_experiment/Scaling:
plot_scaling.py
scaling_graph.png
scaling_results.csv
Screenshot 2026-06-17 at 10.58.46 AM.png

./HalfShift_experiment:
Halfshift_terminal.pdf  halfshift_trace.csv

./Mirrored_experiment:
mirrored_terminal.pdf  mirrored_trace.csv
stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern % cat "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/adjacent_experiment/adjacent_sst.py"
import sst

cpu = sst.Component("cpu", "memHierarchy.standardCPU")

cpu.addParams({
    "clock" : "2GHz",
    "memFreq" : 4,
    "memSize" : "1KiB",
    "verbose" : 0,
    "maxOutstanding" : 16,
    "opCount" : 50000,
    "reqsPerIssue" : 2,
    "write_freq" : 40,
    "read_freq" : 60
})

iface = cpu.setSubComponent(
    "memory",
    "memHierarchy.standardInterface"
)

cache = sst.Component(
    "l1cache",
    "memHierarchy.Cache"
)

cache.addParams({
    "cache_frequency" : "2GHz",
    "cache_size" : "4KB",
    "associativity" : 4,
    "access_latency_cycles" : 5,
    "replacement_policy" : "lru",
    "coherence_protocol" : "MESI",
    "L1" : 1

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})

memctrl = sst.Component(
    "memory",
    "memHierarchy.MemController"
)

memctrl.addParams({
    "clock" : "1GHz",
    "backing" : "none"
})

memory = memctrl.setSubComponent(
    "backend",
    "memHierarchy.simpleMem"
)

memory.addParams({
    "access_time" : "100ns",
    "mem_size" : "512MiB"
})

link_cpu_cache = sst.Link("link_cpu_cache")

link_cpu_cache.connect(
    (iface, "lowlink", "1ns"),
    (cache, "highlink", "1ns")
)

link_cache_mem = sst.Link("link_cache_mem")

link_cache_mem.connect(
    (cache, "lowlink", "5ns"),
    (memctrl, "highlink", "5ns")
)

sst.setStatisticLoadLevel(4)

sst.setStatisticOutput("sst.statOutputCSV")

sst.setStatisticOutputOptions({
    "filepath" : "adjacent_stats.csv",
    "separator" : ",",
})

for comp in [cpu, cache, memctrl]:
    comp.enableAllStatistics()

stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern % head -5 "/Users/stephenrajva/Downloads
/HPC-Upload 2/SST_Stimulator/Communication _Pattern/HalfShift_experiment/halfshift_trace.csv"
echo "-----"
head -5 "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/Mirrored_
experiment/mirrored_trace.csv"
echo "-----"
head -5 "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication _Pattern/adjacent_
experiment/scaling_results.csv"
2,6,400000,0.000145
3,7,400000,0.000332
0,4,400000,0.000338
1,5,400000,0.000179
-----
3,4,400000,0.000088
1,6,400000,0.000144
0,7,400000,0.000171
2,5,400000,0.000133
-----
opCount,Reads,Writes,Cache_GetS,Cache_GetX,PendingCycles
1000,590,410,569,394,16865
5000,2970,2030,2949,2014,57847

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10000,6012,3988,5991,3972,109025
50000,30014,19986,29993,19970,519050
stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern %
stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern % mkdir -p "/Users/stephenrajva/Download
s/HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1"
stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern % find ~/Downloads -name "pattern_mesh_p
pn1.py" 2>/dev/null
find ~/Downloads -name "run_pattern_sweep.sh" 2>/dev/null
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1/pattern_mesh_p
pn1.py
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1/run_pattern_sw
eep.sh
stephenrajva@Stephens-MacBook-Air-4 Communication _Pattern % cd "/Users/stephenrajva/Downloads/HPC-
Upload 2/SST_Stimulator/Communication_Pattern_PPN1"
export PATTERN="adjacent"
export MESH_SHAPE="2x2"
export STATS_DIR="./adjacent/results"
sst pattern_mesh_ppn1.py
=====
Pattern          : adjacent
Topology         : Mesh 2x2
Total nodes      : 4
Total MPI ranks  : 4
Communicating pairs: 2
PPN (ranks/node) : 1    <-- should be 1
Stats CSV        : ./adjacent/results/adjacent_mesh_2x2_ppn1_nodes4_stats.csv
=====
Barrier: ranks 4, loop 5, latency 3.534 us
17848333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=1 messageSize=256 iterations=35
18647933:0:1:EmberEngine:PingPong:generate:65: rank=1 size=4
19447533:0:3:EmberEngine:PingPong:generate:65: rank=3 size=4
PingPong: otherRank 1, total time 69.392 us, loop 35, bufLen 256, latency 0.991 us. bandwidth 0.258
243 GB/s
87240333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=3 messageSize=256 iterations=35
PingPong: otherRank 3, total time 75.320 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237
918 GB/s
Simulation is complete, simulated time: 165.452 us
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 %
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % python3 -c "import sst.ember" 2>/d
ev/null; sst-info ember 2>/dev/null | grep -iE "motif|shift|pingpong|allpingpong|comm" | head -40
Description: Base communicator motif engine.
    motifLog: Sets a file path to a file where motif execution details are written, empty = no
log    []
    motif_count: Sets the number of motifs which will be run in this simulation, default is 1
[1]
    motif%(motif_count)d: Sets the event generator or motif for the engine [ember.EmberPingPo
ngGenerator]
SubComponent 0: 3DAMRMotif
Compiled using file: mpi/motifs/ember3damr.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
    time-Commcreate: Time spent in Commcreate event, (units = "ns") Enable level = 0
SubComponent 1: AllPingPongMotif
Description: Performs a All Ping Pong Motif
Compiled using file: mpi/motifs/emberallpingpong.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
    time-Commcreate: Time spent in Commcreate event, (units = "ns") Enable level = 0
SubComponent 2: AllgatherMotif
Compiled using file: mpi/motifs/emberallgather.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
    time-Commcreate: Time spent in Commcreate event, (units = "ns") Enable level = 0
SubComponent 3: AllgathervMotif
Compiled using file: mpi/motifs/emberallgatherv.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]

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    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
    time-Commcreate: Time spent in Commcreate event, (units = "ns") Enable level = 0
SubComponent 4: AllreduceMotif
    Compiled using file: mpi/motifs/emberallreduce.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
    time-Commcreate: Time spent in Commcreate event, (units = "ns") Enable level = 0
SubComponent 5: AlltoallMotif
    Compiled using file: mpi/motifs/emberalltoall.h
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    time-Commsplit: Time spent in Commsplit event, (units = "ns") Enable level = 0
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % sst-info ember 2>/dev/null | grep
-iE "SubComponent [0-9]+:."
SubComponent 0: 3DAMRMotif
SubComponent 1: AllPingPongMotif
SubComponent 2: AllgatherMotif
SubComponent 3: AllgathervMotif
SubComponent 4: AllreduceMotif
SubComponent 5: AlltoallMotif
SubComponent 6: AlltoallvMotif
SubComponent 7: BFSMotif
SubComponent 8: BarrierMotif
SubComponent 9: BcastMotif
SubComponent 10: BiPingPongMotif
SubComponent 11: CMT1DMotif
SubComponent 12: CMT2DMotif
SubComponent 13: CMT3DMotif
SubComponent 14: CMTCRMotif
SubComponent 15: CommDoubleMotif
SubComponent 16: CommMotif
SubComponent 17: DetailedRingMotif
SubComponent 18: DetailedStreamMotif
SubComponent 19: FFT3DMotif
SubComponent 20: FiniMotif
SubComponent 21: Halo2DMotif
SubComponent 22: Halo2DNBRMotif
SubComponent 23: Halo3D26Motif
SubComponent 24: Halo3DMotif
SubComponent 25: Halo3DSVMotif
SubComponent 26: HotSpotsMotif
SubComponent 27: IncastMotif
SubComponent 28: InitMotif
SubComponent 29: LQCDMotif
SubComponent 30: MsgRateMotif
SubComponent 31: NASLUMotif
SubComponent 32: NtoMMotif
SubComponent 33: NullMotif
SubComponent 34: PingPongMotif
SubComponent 35: RandomMotif
SubComponent 36: ReduceMotif
SubComponent 37: RingMotif
SubComponent 38: SIRIUSTraceMotif
SubComponent 39: ScatterMotif
SubComponent 40: ScattervMotif
SubComponent 41: SendrecvMotif
SubComponent 42: ShmemAddDoubleMotif
SubComponent 43: ShmemAddFloatMotif
SubComponent 44: ShmemAddIntMotif
SubComponent 45: ShmemAddLongMotif
SubComponent 46: ShmemAlltoall32Motif
SubComponent 47: ShmemAlltoall64Motif
SubComponent 48: ShmemAlltoalls32Motif
SubComponent 49: ShmemAlltoalls64Motif
SubComponent 50: ShmemAtomicIncIntMotif
SubComponent 51: ShmemAtomicIncLongMotif
SubComponent 52: ShmemAtomicIncV2IntMotif
SubComponent 53: ShmemAtomicIncV2LongMotif
SubComponent 54: ShmemBarrierAllMotif
SubComponent 55: ShmemBarrierMotif
SubComponent 56: ShmemBroadcast32Motif

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SubComponent 57: ShmemBroadcast64Motif
SubComponent 58: ShmemCollect32Motif
SubComponent 59: ShmemCollect64Motif
SubComponent 60: ShmemCswapDoubleMotif
SubComponent 61: ShmemCswapFloatMotif
SubComponent 62: ShmemCswapIntMotif
SubComponent 63: ShmemCswapLongMotif
SubComponent 64: ShmemFAM_AtomicIncIntMotif
SubComponent 65: ShmemFAM_AtomicIncLongMotif
SubComponent 66: ShmemFAM_CswapIntMotif
SubComponent 67: ShmemFAM_CswapLongMotif
SubComponent 68: ShmemFAM_GathervMotif
SubComponent 69: ShmemFAM_Get2Motif
SubComponent 70: ShmemFAM_PutMotif
SubComponent 71: ShmemFAM_ScattervMotif
SubComponent 72: ShmemFaddDoubleMotif
SubComponent 73: ShmemFaddFloatMotif
SubComponent 74: ShmemFaddIntMotif
SubComponent 75: ShmemFaddLongMotif
SubComponent 76: ShmemFcollect32Motif
SubComponent 77: ShmemFcollect64Motif
SubComponent 78: ShmemGetDoubleMotif
SubComponent 79: ShmemGetFloatMotif
SubComponent 80: ShmemGetIntMotif
SubComponent 81: ShmemGetLongMotif
SubComponent 82: ShmemGetNBIDoubleMotif
SubComponent 83: ShmemGetNBIFloatMotif
SubComponent 84: ShmemGetNBIIntMotif
SubComponent 85: ShmemGetNBILongMotif
SubComponent 86: ShmemGetvDoubleMotif
SubComponent 87: ShmemGetvFloatMotif
SubComponent 88: ShmemGetvIntMotif
SubComponent 89: ShmemGetvLongMotif
SubComponent 90: ShmemHotAtomicIncIntMotif
SubComponent 91: ShmemHotAtomicIncLongMotif
SubComponent 92: ShmemHotAtomicIncV2IntMotif
SubComponent 93: ShmemHotAtomicIncV2LongMotif
SubComponent 94: ShmemNSAtomicIncIntMotif
SubComponent 95: ShmemNSAtomicIncV2IntMotif
SubComponent 96: ShmemPutDoubleMotif
SubComponent 97: ShmemPutFloatMotif
SubComponent 98: ShmemPutIntMotif
SubComponent 99: ShmemPutLongMotif
SubComponent 100: ShmemPutvDoubleMotif
SubComponent 101: ShmemPutvFloatMotif
SubComponent 102: ShmemPutvIntMotif
SubComponent 103: ShmemPutvLongMotif
SubComponent 104: ShmemReductionDoubleMotif
SubComponent 105: ShmemReductionFloatMotif
SubComponent 106: ShmemReductionIntMotif
SubComponent 107: ShmemReductionLongLongMotif
SubComponent 108: ShmemReductionLongMotif
SubComponent 109: ShmemRing2IntMotif
SubComponent 110: ShmemRingIntMotif
SubComponent 111: ShmemSwapIntMotif
SubComponent 112: ShmemSwapLongMotif
SubComponent 113: ShmemTestMotif
SubComponent 114: ShmemWaitIntMotif
SubComponent 115: ShmemWaitLongMotif
SubComponent 116: ShmemWaitUntilIntMotif
SubComponent 117: ShmemWaitUntilLongMotif
SubComponent 118: StopMotif
SubComponent 119: Sweep2DMotif
SubComponent 120: Sweep3DMotif
SubComponent 121: TestMotif
SubComponent 122: TestanyMotif
SubComponent 123: TrafficGenMotif
SubComponent 124: TriCountMotif
SubComponent 125: UnstructuredMotif
SubComponent 126: WaitanyMotif

stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % sst-info ember 2>/dev/null | grep
-A 25 "SubComponent 41: SendrecvMotif"

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SubComponent 41: SendrecvMotif
Description: Performs a Sendrecv Motif
ELI version: 0.9.0
Compiled using file: mpi/motifs/embersendrecv.h
Interface: SST::Ember::EmberGenerator
Parameters (1 total)
    rankmapper: Sets the rank mapper module []
Ports (0 total)
SubComponent Slots (0 total)
Statistics (22 total)
    time-Init: Time spent in Init event, (units = "ns") Enable level = 0
    time-Finalize: Time spent in Finalize event, (units = "ns") Enable level = 0
    time-Rank: Time spent in Rank event, (units = "ns") Enable level = 0
    time-Size: Time spent in Size event, (units = "ns") Enable level = 0
    time-Send: Time spent in Recv event, (units = "ns") Enable level = 0
    time-Recv: Time spent in Recv event, (units = "ns") Enable level = 0
    time-Irecv: Time spent in Irecv event, (units = "ns") Enable level = 0
    time-Isend: Time spent in Isend event, (units = "ns") Enable level = 0
    time-Wait: Time spent in Wait event, (units = "ns") Enable level = 0
    time-Waitall: Time spent in Waitall event, (units = "ns") Enable level = 0
    time-Waitany: Time spent in Waitany event, (units = "ns") Enable level = 0
    time-Sendrecv: Time spent in Sendrecv event, (units = "ns") Enable level = 0
    time-Compute: Time spent in Compute event, (units = "ns") Enable level = 0
    time-Barrier: Time spent in Barrier event, (units = "ns") Enable level = 0
    time-Alltoallv: Time spent in Alltoallv event, (units = "ns") Enable level = 0
    time-Alltoall: Time spent in Alltoall event, (units = "ns") Enable level = 0
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % sst-info ember 2>/dev/null | grep
-A 25 "SubComponent 1: AllPingPongMotif"
SubComponent 1: AllPingPongMotif
Description: Performs a All Ping Pong Motif
ELI version: 0.9.0
Compiled using file: mpi/motifs/emberallpingpong.h
Interface: SST::Ember::EmberGenerator
Parameters (8 total)
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    _jobId: used internally [-1]
    _enginePtr: used internally [-1]
    distribModule: Sets the distribution SST module for compute modeling, default is a constant
distribution of mean 1 [1.0]
    arg.iterations: Sets the number of ping pong operations to perform [1024]
    arg.messageSize: Sets the message size of the ping pong operation [128]
    arg.computetime: Sets the time spent computing some values [1000]
Ports (0 total)
SubComponent Slots (0 total)
Statistics (21 total)
    time-Init: Time spent in Init event, (units = "ns") Enable level = 0
    time-Finalize: Time spent in Finalize event, (units = "ns") Enable level = 0
    time-Rank: Time spent in Rank event, (units = "ns") Enable level = 0
    time-Size: Time spent in Size event, (units = "ns") Enable level = 0
    time-Send: Time spent in Recv event, (units = "ns") Enable level = 0
    time-Recv: Time spent in Recv event, (units = "ns") Enable level = 0
    time-Irecv: Time spent in Irecv event, (units = "ns") Enable level = 0
    time-Isend: Time spent in Isend event, (units = "ns") Enable level = 0
    time-Wait: Time spent in Wait event, (units = "ns") Enable level = 0
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % sst-info ember 2>/dev/null | grep
-A 20 "SubComponent 123: TrafficGenMotif"
echo "=====
sst-info ember 2>/dev/null | grep -A 20 "SubComponent 16: CommMotif"
SubComponent 123: TrafficGenMotif
Description: Models network traffic
ELI version: 0.9.0
Compiled using file: mpi/motifs/emberTrafficGen.h
Interface: SST::Ember::EmberGenerator
Parameters (9 total)
    primary: is this a primary motif [0]
    _motifNum: used internally [-1]
    _jobId: used internally [-1]
    _enginePtr: used internally [-1]
    distribModule: Sets the distribution SST module for compute modeling, default is a constant
distribution of mean 1 [1.0]
    arg.messageSize: Sets the size of exchange [1]

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[illegible]

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stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 %
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % cd "/Users/stephenrajva/Downloads/
HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1"
export PATTERN="halfshift"
export MESH_SHAPE="4x4"
export STATS_DIR="./halfshift/results"
mkdir -p ./halfshift/results
sst pattern_mesh_ppn1.py
=====
Pattern          : halfshift
Topology         : Mesh 4x4 (16 nodes, PPN=1)
Exchange        : rank 0 <-> rank 8
Representative hops: 2 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_4x4_ppn1_nodes16_stats.csv
=====
180000:0:8:EmberEngine:PingPong:generate:65: rank=8 size=16
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=8 messageSize=256 iterations=35
PingPong: otherRank 8, total time 75.319 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237
921 GB/s
Simulation is complete, simulated time: 81.7851 us
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % cd "/Users/stephenrajva/Downloads/
HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1"

for P in adjacent halfshift mirrored; do
  mkdir -p "$P/results"
  for S in 2x2 4x4 8x8 16x16 32x32; do
    echo "=== $P on mesh $S ==="
    export PATTERN="$P"
    export MESH_SHAPE="$S"
    export STATS_DIR="$P/results"
    sst pattern_mesh_ppn1.py
  done
done
=== adjacent on mesh 2x2 ===
=====
Pattern          : adjacent
Topology         : Mesh 2x2 (4 nodes, PPN=1)
Exchange        : rank 0 <-> rank 1
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./adjacent/results/adjacent_mesh_2x2_ppn1_nodes4_stats.csv
=====
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=4
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 69.271 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
694 GB/s
Simulation is complete, simulated time: 72.0203 us
=== adjacent on mesh 4x4 ===
=====
Pattern          : adjacent
Topology         : Mesh 4x4 (16 nodes, PPN=1)
Exchange        : rank 0 <-> rank 1
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./adjacent/results/adjacent_mesh_4x4_ppn1_nodes16_stats.csv
=====
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=16
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 69.284 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
646 GB/s
Simulation is complete, simulated time: 74.1955 us
=== adjacent on mesh 8x8 ===
=====
Pattern          : adjacent
Topology         : Mesh 8x8 (64 nodes, PPN=1)
Exchange        : rank 0 <-> rank 1
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./adjacent/results/adjacent_mesh_8x8_ppn1_nodes64_stats.csv
=====
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=64
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64

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rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 69.271 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
694 GB/s
Simulation is complete, simulated time: 76.7139 us
=== adjacent on mesh 16x16 ===
=====
Pattern          : adjacent
Topology         : Mesh 16x16 (256 nodes, PPN=1)
Exchange        : rank 0 <-> rank 1
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./adjacent/results/adjacent_mesh_16x16_ppn1_nodes256_stats.csv
=====
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=256
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=256
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 69.271 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
694 GB/s
Simulation is complete, simulated time: 81.2947 us
=== adjacent on mesh 32x32 ===
=====
Pattern          : adjacent
Topology         : Mesh 32x32 (1024 nodes, PPN=1)
Exchange        : rank 0 <-> rank 1
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./adjacent/results/adjacent_mesh_32x32_ppn1_nodes1024_stats.csv
=====
180000:0:1:EmberEngine:PingPong:generate:65: rank=1 size=1024
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=1024
rank2=1 messageSize=256 iterations=35
PingPong: otherRank 1, total time 69.271 us, loop 35, bufLen 256, latency 0.990 us. bandwidth 0.258
694 GB/s
Simulation is complete, simulated time: 88.8643 us
=== halfshift on mesh 2x2 ===
=====
Pattern          : halfshift
Topology         : Mesh 2x2 (4 nodes, PPN=1)
Exchange        : rank 0 <-> rank 2
Representative hops: 1 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_2x2_ppn1_nodes4_stats.csv
=====
180000:0:2:EmberEngine:PingPong:generate:65: rank=2 size=4
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=2 messageSize=256 iterations=35
PingPong: otherRank 2, total time 69.361 us, loop 35, bufLen 256, latency 0.991 us. bandwidth 0.258
358 GB/s
Simulation is complete, simulated time: 71.9603 us
=== halfshift on mesh 4x4 ===
=====
Pattern          : halfshift
Topology         : Mesh 4x4 (16 nodes, PPN=1)
Exchange        : rank 0 <-> rank 8
Representative hops: 2 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_4x4_ppn1_nodes16_stats.csv
=====
180000:0:8:EmberEngine:PingPong:generate:65: rank=8 size=16
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=8 messageSize=256 iterations=35
PingPong: otherRank 8, total time 75.319 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237
921 GB/s
Simulation is complete, simulated time: 81.7851 us
=== halfshift on mesh 8x8 ===
=====
Pattern          : halfshift
Topology         : Mesh 8x8 (64 nodes, PPN=1)
Exchange        : rank 0 <-> rank 32
Representative hops: 4 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_8x8_ppn1_nodes64_stats.csv
=====
180000:0:32:EmberEngine:PingPong:generate:65: rank=32 size=64
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=32 messageSize=256 iterations=35
PingPong: otherRank 32, total time 87.415 us, loop 35, bufLen 256, latency 1.249 us. bandwidth 0.20

```

```

4999 GB/s
Simulation is complete, simulated time: 98.9163 us
=== halfshift on mesh 16x16 ===
=====
Pattern          : halfshift
Topology         : Mesh 16x16 (256 nodes, PPN=1)
Exchange        : rank 0 <-> rank 128
Representative hops: 8 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_16x16_ppn1_nodes256_stats.csv
=====
180000:0:128:EmberEngine:PingPong:generate:65: rank=128 size=256
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=256
rank2=128 messageSize=256 iterations=35
PingPong: otherRank 128, total time 111.607 us, loop 35, bufLen 256, latency 1.594 us. bandwidth 0.
160563 GB/s
Simulation is complete, simulated time: 130.758 us
=== halfshift on mesh 32x32 ===
=====
Pattern          : halfshift
Topology         : Mesh 32x32 (1024 nodes, PPN=1)
Exchange        : rank 0 <-> rank 512
Representative hops: 16 (Manhattan dist on mesh)
Stats CSV       : ./halfshift/results/halfshift_mesh_32x32_ppn1_nodes1024_stats.csv
=====
180000:0:512:EmberEngine:PingPong:generate:65: rank=512 size=1024
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=1024
rank2=512 messageSize=256 iterations=35
PingPong: otherRank 512, total time 159.991 us, loop 35, bufLen 256, latency 2.286 us. bandwidth 0.
112006 GB/s
Simulation is complete, simulated time: 191.163 us
=== mirrored on mesh 2x2 ===
=====
Pattern          : mirrored
Topology         : Mesh 2x2 (4 nodes, PPN=1)
Exchange        : rank 0 <-> rank 3
Representative hops: 2 (Manhattan dist on mesh)
Stats CSV       : ./mirrored/results/mirrored_mesh_2x2_ppn1_nodes4_stats.csv
=====
180000:0:3:EmberEngine:PingPong:generate:65: rank=3 size=4
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=3 messageSize=256 iterations=35
PingPong: otherRank 3, total time 75.319 us, loop 35, bufLen 256, latency 1.076 us. bandwidth 0.237
921 GB/s
Simulation is complete, simulated time: 78.3915 us
=== mirrored on mesh 4x4 ===
=====
Pattern          : mirrored
Topology         : Mesh 4x4 (16 nodes, PPN=1)
Exchange        : rank 0 <-> rank 15
Representative hops: 6 (Manhattan dist on mesh)
Stats CSV       : ./mirrored/results/mirrored_mesh_4x4_ppn1_nodes16_stats.csv
=====
180000:0:15:EmberEngine:PingPong:generate:65: rank=15 size=16
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=15 messageSize=256 iterations=35
PingPong: otherRank 15, total time 99.511 us, loop 35, bufLen 256, latency 1.422 us. bandwidth 0.18
0081 GB/s
Simulation is complete, simulated time: 106.302 us
=== mirrored on mesh 8x8 ===
=====
Pattern          : mirrored
Topology         : Mesh 8x8 (64 nodes, PPN=1)
Exchange        : rank 0 <-> rank 63
Representative hops: 14 (Manhattan dist on mesh)
Stats CSV       : ./mirrored/results/mirrored_mesh_8x8_ppn1_nodes64_stats.csv
=====
180000:0:63:EmberEngine:PingPong:generate:65: rank=63 size=64
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=63 messageSize=256 iterations=35
PingPong: otherRank 63, total time 147.895 us, loop 35, bufLen 256, latency 2.113 us. bandwidth 0.1
21167 GB/s
Simulation is complete, simulated time: 159.03 us

```

=== mirrored on mesh 16x16 ===

```
=====
Pattern      : mirrored
Topology     : Mesh 16x16 (256 nodes, PPN=1)
Exchange     : rank 0 <-> rank 255
Representative hops: 30 (Manhattan dist on mesh)
Stats CSV    : ./mirrored/results/mirrored_mesh_16x16_ppn1_nodes256_stats.csv
=====
```

```
180000:0:255:EmberEngine:PingPong:generate:65: rank=255 size=256
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=256
rank2=255 messageSize=256 iterations=35
PingPong: otherRank 255, total time 244.663 us, loop 35, bufLen 256, latency 3.495 us. bandwidth 0.
073244 GB/s
Simulation is complete, simulated time: 262.065 us
=== mirrored on mesh 32x32 ===
```

```
=====
Pattern      : mirrored
Topology     : Mesh 32x32 (1024 nodes, PPN=1)
Exchange     : rank 0 <-> rank 1023
Representative hops: 62 (Manhattan dist on mesh)
Stats CSV    : ./mirrored/results/mirrored_mesh_32x32_ppn1_nodes1024_stats.csv
=====
```

```
180000:0:1023:EmberEngine:PingPong:generate:65: rank=1023 size=1024
180000:0:0:EmberEngine:PingPong:generate:65: rank=0 size=1024
rank2=1023 messageSize=256 iterations=35
PingPong: otherRank 1023, total time 438.199 us, loop 35, bufLen 256, latency 6.260 us. bandwidth 0.
.040895 GB/s
Simulation is complete, simulated time: 464.857 us
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 %
stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 % cd "/Users/stephenrajva/Downloads/
HPC-Upload 2/SST_Stimulator/Communication_Pattern_PPN1"
```

python3 analyze_patterns.py

Pattern	Size	Nodes	Hops	Latency(ns)
---------	------	-------	------	-------------

adjacent	2x2	4	1	198.51
adjacent	4x4	16	1	230.97
adjacent	8x8	64	1	386.15
adjacent	16x16	256	1	764.17
adjacent	32x32	1024	1	1492.46
halfshift	2x2	4	1	198.17
halfshift	4x4	16	2	291.80
halfshift	8x8	64	4	478.78
halfshift	16x16	256	8	837.15
halfshift	32x32	1024	16	1535.34
mirrored	2x2	4	2	278.12
mirrored	4x4	16	6	533.39
mirrored	8x8	64	14	787.30
mirrored	16x16	256	30	1066.58
mirrored	32x32	1024	62	1666.84

Saved: pattern_latency_vs_size.png

Saved: pattern_hops_vs_size.png

stephenrajva@Stephens-MacBook-Air-4 Communication_Pattern_PPN1 %